

Skill Arena - Phase 6 Document 2

AI Systems, Smart Matchmaking, Personalization Engine & Intelligent Platform Services Architecture

Document Purpose

Define the future AI architecture powering personalization, intelligent matchmaking, recommendations and operational intelligence.

Business Objective

Increase engagement, retention, fairness and operational efficiency using intelligent platform services.

AI Vision

Use AI to improve player experience while preserving fairness, transparency and competitive integrity.

Core AI Components

Personalization Engine, Smart Matchmaking Engine, Recommendation Engine, AI Analytics Engine and AI Operations Layer.

Personalization Engine

Adapts dashboards, recommendations, events and progression opportunities to individual players.

Player Profiles

Build behavioural models using gameplay, progression, engagement and activity patterns.

Recommendation Engine

Suggest games, tournaments, challenges, rewards and progression opportunities.

Smart Matchmaking

Enhance matchmaking using behavioural insights, engagement patterns and competitive balance metrics.

Player Retention Models

Identify churn risks and recommend engagement strategies.

Difficulty Optimization

Recommend challenge difficulty ranges based on player performance and progression.

AI Tournament Insights

Suggest tournament participation opportunities and qualification targets.

AI Reward Optimization

Evaluate reward structures and seasonal incentives.

AI Moderation Services

Assist moderators by identifying suspicious behaviour, abuse patterns and enforcement candidates.

AI Fraud Signals

Support fraud detection systems through anomaly analysis and behavioural modelling.

AI Analytics Platform

Generate insights for product teams, finance teams and operational staff.

Predictive Analytics

Forecast growth, retention, treasury requirements and player engagement trends.

AI Operations Layer

Support automated operational recommendations and monitoring.

Ethical AI Principles

Fairness, transparency, explainability and player protection.

Privacy Requirements

AI systems operate within platform privacy, security and compliance requirements.

Human Oversight

Critical financial, security and enforcement decisions require human review.

Model Monitoring

Track performance, drift, accuracy and operational impact of AI systems.

Scalability Requirements

Support global player populations and future platform expansion.

Security Requirements

Protect AI models, training data and decision systems from manipulation.

Audit Requirements

Record AI-driven recommendations, risk signals and operational decisions.

Future Expansion

Foundation for AI coaches, AI training systems and advanced player assistance.

Acceptance Criteria

AI services improve engagement and operational efficiency while maintaining trust and fairness.